

4 DEC 2025

HORSE (Hospital Operations and Routine Staffing Efficiency)

Proactively Optimizing HR Allocation with Agentic AI to Minimize Labor Costs and Staff Burnout

Team 6

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Hospitals rely on manual, rigid staffing processes that can't keep up with real-time patient demand.

This creates constant operational firefighting, unpredictable coverage, and avoidable dependency on premium labor.

Current Hospital Staffing Pain Points

- The current staffing workflow is slow, labor-intensive, and reactive.
- 13.4 hours to build each schedule (per person per 4-week cycle)
- 40.3 hours spent fixing avoidable issues
- No real-time adjustment → constant last-minute scrambling

PAIN POINTS

The Cost of Inefficient Hospital Staffing Today

+\$10M - \$13.5M

Annual cost of travel nurses

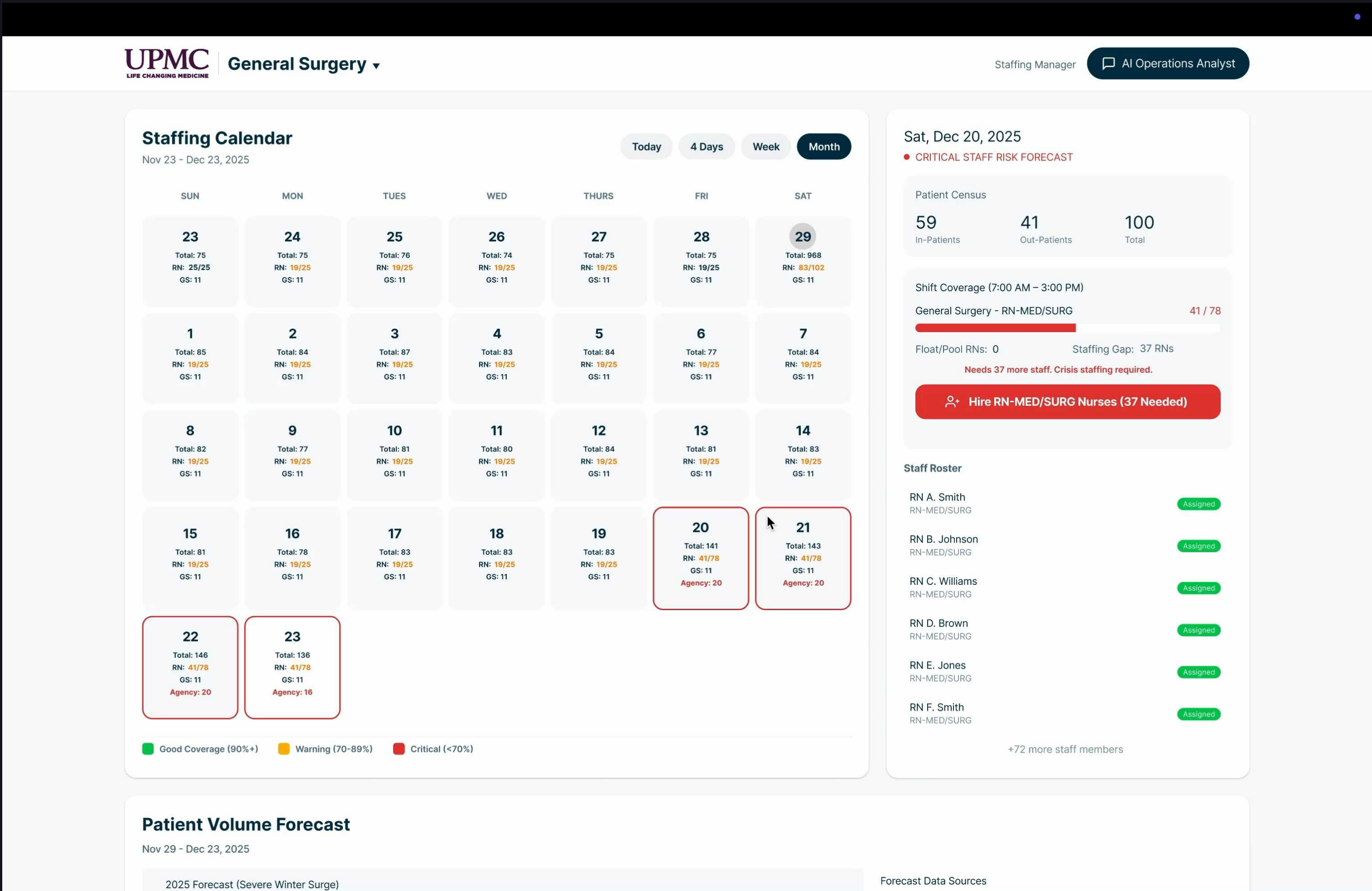
+\$400K - \$650K

Annual cost of overtime & premium pay

The Product

The AI Agent autonomously generates a routine staffing schedule.

- predicts future staffing needs
- optimizes resource allocation by prioritizing the lowest-cost internal float nurses
- directly executes external contingent labor (travel nurses) sourcing through automated competitive bidding.



Real Time Updates

The AI Agent autonomously generates a routine staffing schedule and keeps track real time changes in:

- 1. Patient in-take/changes
- 2. Staff needs/shortage

Sat, Dec 20, 2025

● CRITICAL STAFF RISK FORECAST

Patient Census

59

In-Patients

41

Out-Patients

100

Total

Shift Coverage (7:00 AM – 3:00 PM)

General Surgery - RN-MED/SURG

41 / 78

Float/Pool RNs: 0

Staffing Gap: 37 RNs

Needs 37 more staff. Crisis staffing required.



Hire RN-MED/SURG Nurses (37 Needed)

DEMO: SOLUTION 1

Real Time Updates

The AI Agent autonomously generates a routine staffing schedule and keeps track real time changes in:

- 1. Patient in-take/changes
- 2. Staff needs/shortage

Today

4 Days

Week

Month

AI Operations Analyst

RN-MED/SURG Hiring Options

December 20, 2025 - Staffing Gap: 37 RNs

Selected Nurses

0

Total Cost (4 days)

\$0.0K

Savings vs Market

\$0.0K

AI-Powered Quick Hire

Let AI automatically select the most cost-effective nurses for you. Prioritizes internal staff (45% lower cost) and optimizes for quality and availability.

Auto-Select 37 Best Nurses

Available Nurses of Live Bidding (Sorted by Cost)

☐

Internal RN - Pool 1

BEST VALUE

INTERNAL

Type: Internal Pool

Experience: 8 years

Specialty: MED/SURG

Rating: ★ 4.8

\$45/hr

\$1440 for 4 days

☐

Internal RN - Pool 2

INTERNAL

Type: Internal Pool

Experience: 5 years

Specialty: MED/SURG

Rating: ★ 4.6

\$48/hr

\$1536 for 4 days

☐

HealthForce Solutions - RN A

Type: Agency

Experience: 6 years

Specialty: MED/SURG

Rating: ★ 4.7

\$82/hr

\$2624 for 4 days

☐

HealthForce Solutions - RN B

Type: Agency

Experience: 4 years

Specialty: MED/SURG

Rating: ★ 4.5

\$82/hr

\$2624 for 4 days

☐

Elite Medical Staffing - RN A

Type: Agency

Experience: 7 years

Specialty: MED/SURG

Rating: ★ 4.6

\$85/hr

\$2720 for 4 days

Selected: 0 / 37 needed

Cancel

Confirm 0 Nurses

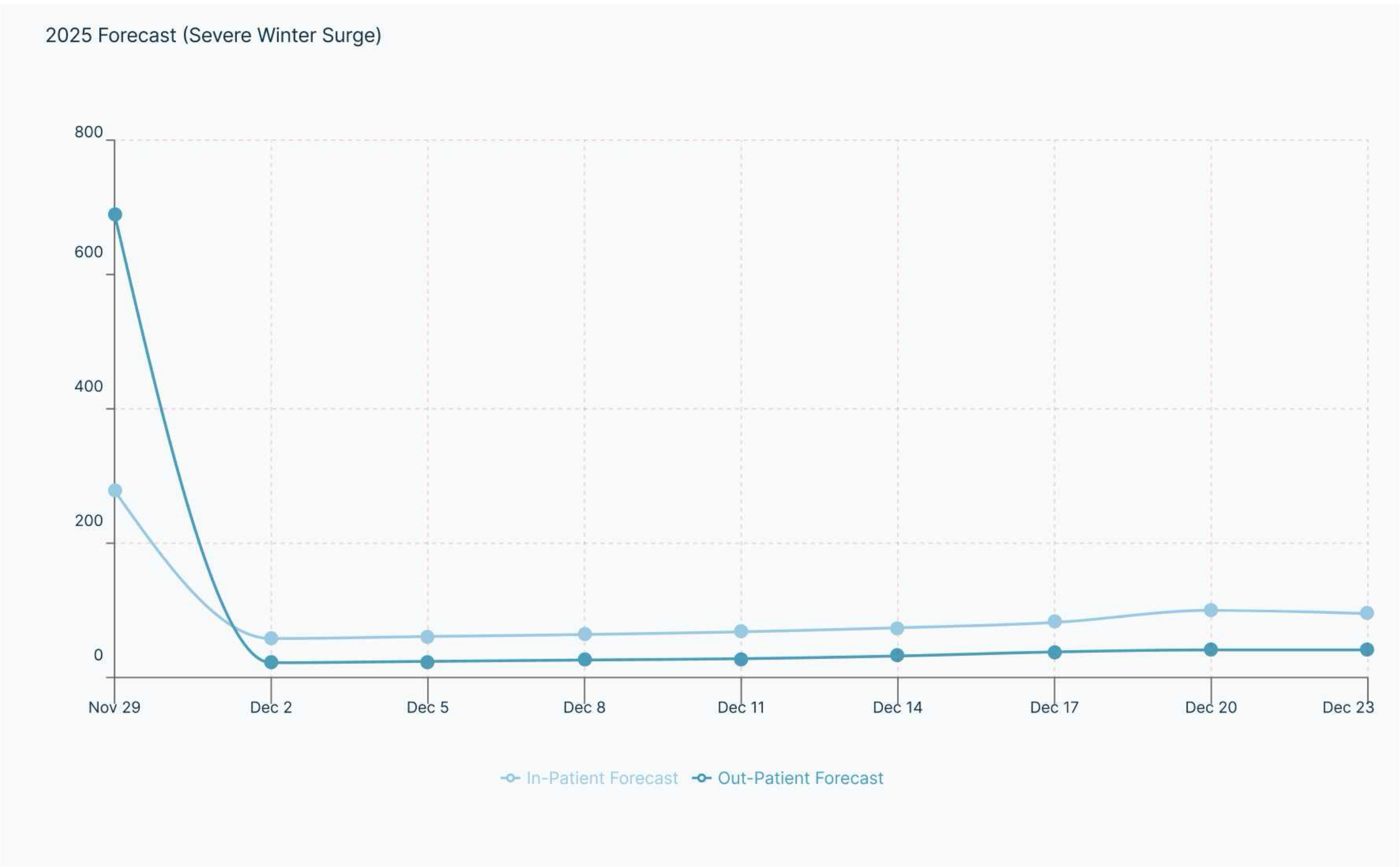
Future Forecast: Staff Shortage

The AI Agent notes that there's a future staff shortage based on predictive model and initiates

- 1. internal float nurses allocation
- 2. external agency nurse bidding

Patient Volume Forecast

Nov 29 - Dec 23, 2025



Future Forecast: Staff Shortage

The AI Agent notes that there’s a future staff shortage based on predictive model and initiates

- 1. internal float nurses allocation
- 2. external agency nurse bidding

RN-MED/SURG Hiring Options

December 20, 2025 - Staffing Gap: 37 RNs

Selected Nurses

7

Total Cost (4 days)

\$16.5K

Savings vs Market

\$3.2K

AI-Powered Quick Hire

Let AI automatically select the most cost-effective nurses for you. Prioritizes internal staff (45% lower cost) and optimizes for quality and availability.

Auto-Select 37 Best Nurses

Available Nurses (Sorted by Cost)

Internal RN - Pool 1

BEST VALUE

INTERNAL

Type: Internal Pool Experience: 8 years Specialty: MED/SURG Rating: ★ 4.8

\$45/hr

\$1440 for 4 days

Internal RN - Pool 2

INTERNAL

Type: Internal Pool Experience: 5 years Specialty: MED/SURG Rating: ★ 4.6

\$48/hr

\$1536 for 4 days

HealthForce Solutions - RN A

Type: Agency Experience: 6 years Specialty: MED/SURG Rating: ★ 4.7

\$82/hr

\$2624 for 4 days

HealthForce Solutions - RN B

Type: Agency Experience: 4 years Specialty: MED/SURG Rating: ★ 4.5

\$82/hr

\$2624 for 4 days

Elite Medical Staffing - RN A

Type: Agency Experience: 7 years Specialty: MED/SURG Rating: ★ 4.6

\$85/hr

\$2720 for 4 days

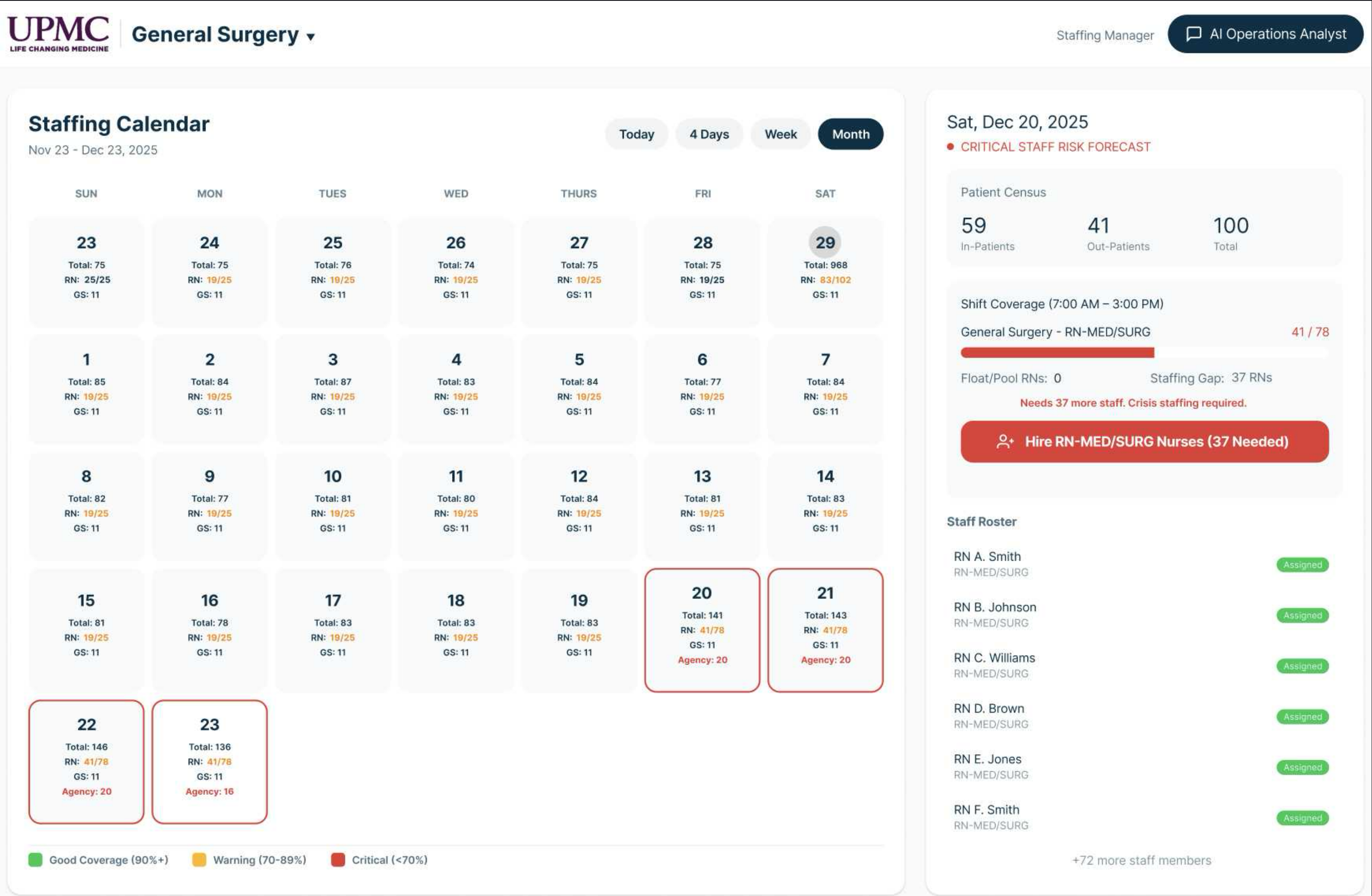
Selected: 7 / 37 needed

Cancel

Confirm 7 Nurses

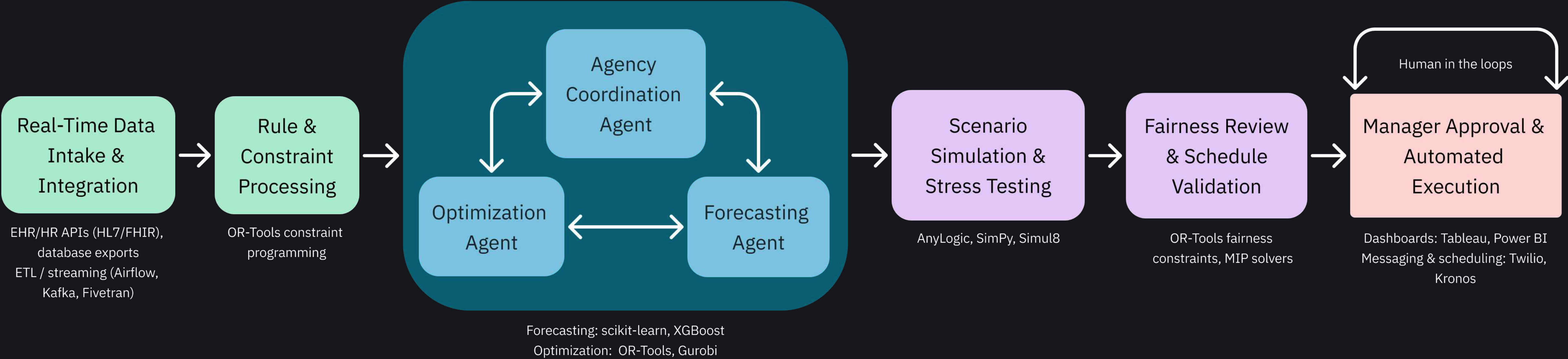
Future Forecast: Calendar View

The AI Agent takes past patient data from the time period (ex. Dec 2024) and inputs current updates (from Dec 2025) and generates predictive model.



TECHNICAL FEASIBILITY

Existing Data and Tools



\$ Total Investment

Costs

Development	\$970K - \$1.4M
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Per Hospital Setup	\$60K - \$100K
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Annual SaaS Fee	\$120K - \$200K
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\$ Hospital Savings/Year

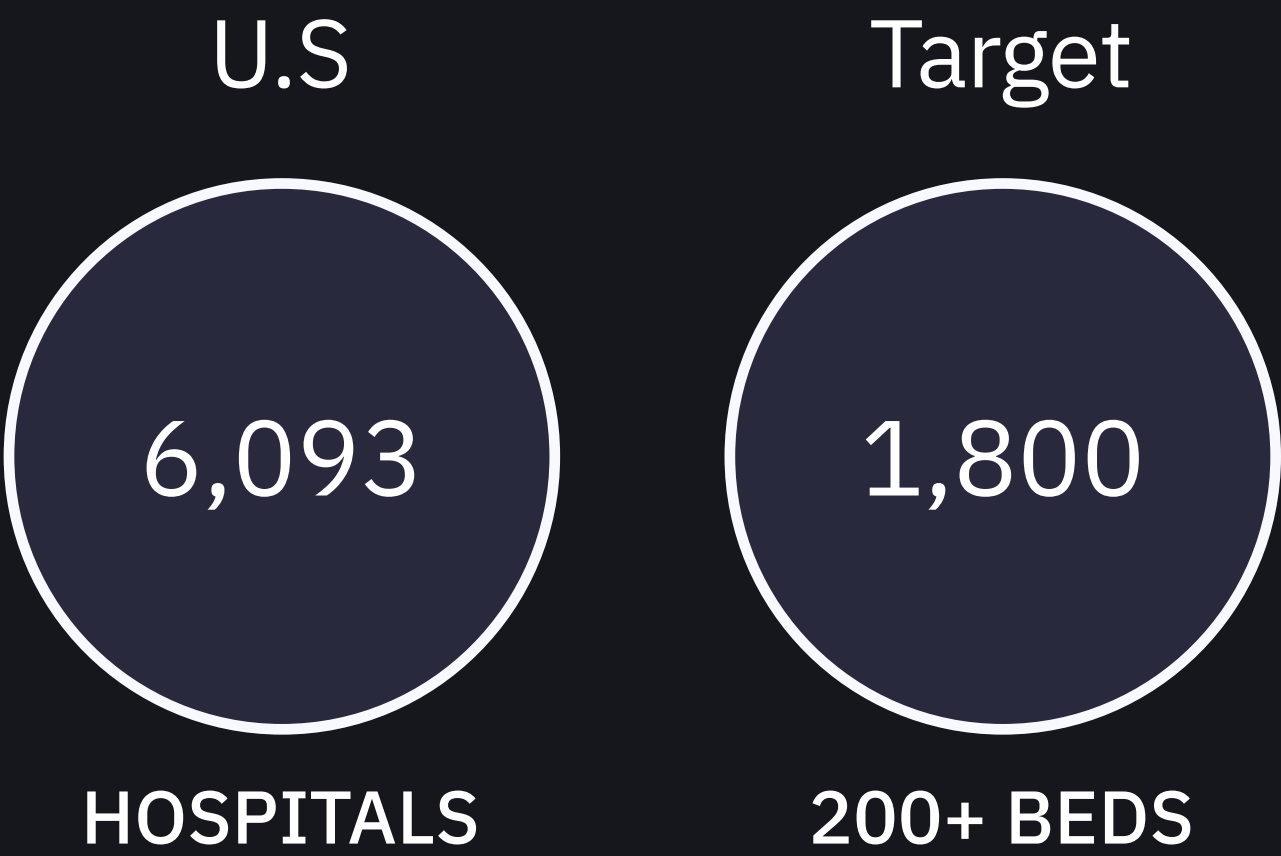
Returns

Travel Nurse Reduction	\$1.2M - \$2.0M
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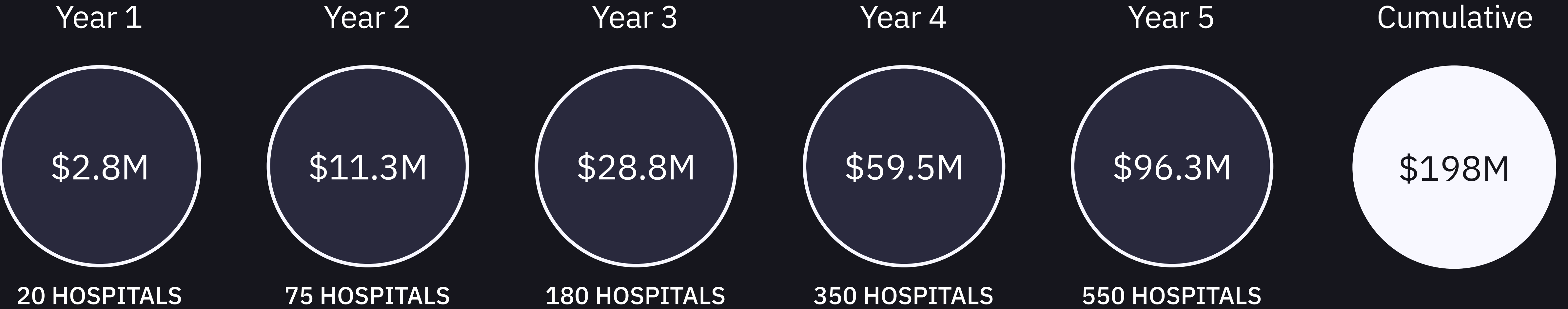
Turnover + Overtime Cuts	\$400K - \$650K
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Total Annual Savings	\$1.6M - \$2.65M
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Total Addressable Market



5-Year Revenue Projection



Let HORSE be your AI staffing agent, automatically predicting demand, filling shifts, and keeping your hospital safely staffed.

Thank you